

Foundation Mathematics 1017SCG
Week 5 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$\begin{aligned}5x + 4 &= 19 \\5x + 4 - 4 &= 19 - 4 \\5x &= 15 \\\frac{5x}{5} &= \frac{15}{5} \\x &= 3\end{aligned}$$

(b)

$$\begin{aligned}7x - 2 &= 33 \\7x - 2 + 2 &= 33 + 2 \\7x &= 35 \\\frac{7x}{7} &= \frac{35}{7} \\x &= 5\end{aligned}$$

(c)

$$\begin{aligned}3x + 9 &= -3 \\3x + 9 - 9 &= -3 - 9 \\3x &= -12 \\\frac{3x}{3} &= \frac{-12}{3} \\x &= -4\end{aligned}$$

(d)

$$\begin{aligned}-2x + 6 &= 26 \\-2x + 6 - 6 &= 26 - 6 \\-2x &= 20 \\\frac{-2x}{-2} &= \frac{20}{-2} \\x &= -10\end{aligned}$$

(e)

$$\begin{aligned}\frac{x}{5} + 2 &= 7 \\\frac{x}{5} + 2 - 2 &= 7 - 2 \\\frac{x}{5} &= 5 \\\frac{x}{5} \times 5 &= 5 \times 5 \\x &= 25\end{aligned}$$

(f)

$$\begin{aligned}\frac{2x}{3} - 4 &= 0 \\ \frac{2x}{3} - 4 + 4 &= 0 + 4 \\ \frac{2x}{3} &= 4 \\ \frac{2x}{3} \times 3 &= 4 \times 3 \\ 2x &= 12 \\ \frac{2x}{2} &= \frac{12}{2} \\ x &= 6\end{aligned}$$

(g)

$$\begin{aligned}6n + 8 &= 44 \\ 6n + 8 - 8 &= 44 - 8 \\ 6n &= 36 \\ \frac{6n}{6} &= \frac{36}{6} \\ n &= 6\end{aligned}$$

(h)

$$\begin{aligned}5(3m - 1) &= 70 \\ \frac{5(3m - 1)}{5} &= \frac{70}{5} \\ 3m - 1 &= 14 \\ 3m - 1 + 1 &= 14 + 1 \\ 3m &= 15 \\ \frac{3m}{3} &= \frac{15}{3} \\ m &= 5\end{aligned}$$

(i)

$$\begin{aligned}-18 - 3h &= 5 \\ -18 + 18 - 3h &= 5 + 18 \\ -3h &= 23 \\ \frac{-3h}{-3} &= \frac{23}{-3} \\ h &= -\frac{23}{3}\end{aligned}$$

(j)

$$\begin{aligned}4x + 7 &= 8 - x \\4x + x + 7 &= 8 - x + x \\5x + 7 &= 8 \\5x + 7 - 7 &= 8 - 7 \\5x &= 1 \\\frac{5x}{5} &= \frac{1}{5} \\x &= \frac{1}{5}\end{aligned}$$

(k)

$$\begin{aligned}2(4y + 1) &= 5(3 - y) \\8y + 2 &= 15 - 5y \\8y + 5y + 2 &= 15 - 5y + 5y \\13y + 2 &= 15 \\13y + 2 - 2 &= 15 - 2 \\13y &= 13 \\\frac{13y}{13} &= \frac{13}{13} \\y &= 1\end{aligned}$$

(l)

$$\begin{aligned}7(5 - c) &= 3(c + 1) - 10 \\35 - 7c &= 3c + 3 - 10 \\35 - 10c &= -7 \\35 - 35 - 10c &= -7 - 35 \\-10c &= -42 \\\frac{-10c}{-10} &= \frac{-42}{-10} \\c &= \frac{21}{5}\end{aligned}$$

2. Throughout this question, factoring has been used to solve some of the quadratic equations. However, the quadratic formula could also have been used to solve the quadratics.

(a)

$$\begin{aligned}x^2 + 5x &= 0 \\x(x + 5) &= 0 \\x = 0, x &= -5\end{aligned}$$

(b)

$$\begin{aligned}x^2 - 4x &= 0 \\x(x - 4) &= 0 \\x = 0, x &= 4\end{aligned}$$

- (c) Note that there are a few different ways that this question could be solved. As there is no linear term in the quadratic, it could be solved by rearranging (which is not usually possible for a quadratic equation).

$$\begin{aligned}x^2 - 9 &= 0 \\x^2 - 3^2 &= 0 \\(x + 3)(x - 3) &= 0 \\x = -3, x = 3\end{aligned}$$

(d)

$$\begin{aligned}x^2 - 13x + 42 &= 0 \\(x - 6)(x - 7) &= 0 \\x = 6, x = 7\end{aligned}$$

(e)

$$\begin{aligned}3x^2 + x - 2 &= 0 \\(3x - 2)(x + 1) &= 0 \\x = \frac{2}{3}, x = -1\end{aligned}$$

- (f) To solve this quadratic, either by factoring or the quadratic formula, it is important that it is first written in standard form. The standard form of a quadratic is $ax^2 + bx + c = 0$ where a , b , and c are constants.

$$\begin{aligned}2x^2 &= 5x + 3 \\2x^2 - 5x - 3 &= 0 \\(2x + 1)(x - 3) &= 0 \\x = -\frac{1}{2}, x = 3\end{aligned}$$

(g)

$$\begin{aligned}2x^2 + 4x - 6 &= 0 \\2(x^2 + 2x - 3) &= 0 \\\frac{2(x^2 + 2x - 3)}{2} &= \frac{0}{2} \\x^2 + 2x - 3 &= 0 \\(x + 3)(x - 1) &= 0 \\x = -3, x = 1\end{aligned}$$

(h)

$$\begin{aligned}x^2 &= 7x - 4 \\x^2 - 7x + 4 &= 0\end{aligned}$$

This quadratic does not factorise easily. Therefore it is necessary to use the quadratic formula.

$$(a = 1, b = -7, c = 4)$$

$$\begin{aligned}
 x &= \frac{7 \pm \sqrt{(-7)^2 - 4 \times 1 \times 4}}{2 \times 1} \\
 &= \frac{7 \pm \sqrt{33}}{2}
 \end{aligned}$$

As the question asks for an exact answer, leave your answer in surd form. If the question did not ask for an exact answer, you could use your scientific calculator to find an approximate values for x .

(i)

$$\begin{aligned}
 2x^2 - 9x + 5 &= 0 \\
 (a = 2, \ b = -9, \ c = 5) \\
 x &= \frac{9 \pm \sqrt{(-9)^2 - 4 \times 2 \times 5}}{2 \times 2} \\
 &= \frac{9 \pm \sqrt{41}}{4}
 \end{aligned}$$

(j)

$$\begin{aligned}
 5x^2 - 3x - 1 &= 0 \\
 (a = 5, \ b = -3, \ c = -1) \\
 x &= \frac{3 \pm \sqrt{(-3)^2 - 4 \times 5 \times -1}}{2 \times 5} \\
 &= \frac{3 \pm \sqrt{29}}{10}
 \end{aligned}$$

3. (a)

$$\begin{aligned}
 \log_2(x) &= 4 \\
 x &= 2^4 \\
 x &= 16
 \end{aligned}$$

(b)

$$\begin{aligned}
 \log_5(x) &= 3 \\
 x &= 5^3 \\
 x &= 125
 \end{aligned}$$

(c)

$$\begin{aligned}
 6 \log_4(x) &= 12 \\
 \log_4(x) &= 2 \\
 x &= 4^2 \\
 x &= 16
 \end{aligned}$$

(d)

$$\begin{aligned}
 \log_3(x) + 7 &= 10 \\
 \log_3(x) &= 3 \\
 x &= 3^3 \\
 x &= 27
 \end{aligned}$$

(e)

$$\log_{10}(5x) = 2$$

$$5x = 10^2$$

$$5x = 100$$

$$x = 20$$

(f)

$$4 \log_7(9x) = 8$$

$$\log_7(9x) = 2$$

$$9x = 7^2$$

$$9x = 49$$

$$x = \frac{49}{9}$$

(g)

$$8 \log_2(4x) = 24$$

$$\log_2(4x) = 3$$

$$4x = 2^3$$

$$4x = 8$$

$$x = 2$$

(h)

$$5 \log_3(6x - 1) = 10$$

$$\log_3(6x - 1) = 2$$

$$6x - 1 = 3^2$$

$$6x - 1 = 9$$

$$6x = 10$$

$$x = \frac{10}{6}$$

$$x = \frac{5}{3}$$

(i)

$$9 \log_5(3x + 2) + 7 = 25$$

$$9 \log_5(3x + 2) = 18$$

$$\log_5(3x + 2) = 2$$

$$3x + 2 = 5^2$$

$$3x + 2 = 25$$

$$3x = 23$$

$$x = \frac{23}{3}$$

(j)

$$3 \log_4(5x + 8) - 4 = 2$$

$$3 \log_4(5x + 8) = 6$$

$$\log_4(5x + 8) = 2$$

$$5x + 8 = 4^2$$

$$5x + 8 = 16$$

$$5x = 8$$

$$x = \frac{8}{5}$$

4. (a)

$$2^x = 12$$

This question could be solved with logarithms of any base. The obvious choices are base 2, base 10 and base e . The solutions for base 2 and base 10 are shown below.

Using base 2 logarithms:

$$2^x = 12$$

$$\log_2(2^x) = \log_2(12)$$

$$x \times \log_2(2) = \log_2(12)$$

$$x \times 1 = \log_2(12)$$

$$x = \log_2(12)$$

However, the question asked us to round our answer to two decimal places. To do this, we will need to use the change of base rule.

$$\begin{aligned} x &= \log_2(12) \\ &= \frac{\log_{10}(12)}{\log_{10}(2)} \\ &= 3.58 \end{aligned}$$

Using base 10 logarithms

$$2^x = 12$$

$$\log_{10}(2^x) = \log_{10}(12)$$

$$x \times \log_{10}(2) = \log_{10}(12)$$

$$x = \frac{\log_{10}(12)}{\log_{10}(2)}$$

$$x = 3.58$$

(b)

$$5^x = 30$$

$$\log_{10}(5^x) = \log_{10}(30)$$

$$x \times \log_{10}(5) = \log_{10}(30)$$

$$x = \frac{\log_{10}(30)}{\log_{10}(5)}$$

$$x = 2.11$$

(c)

$$\begin{aligned}9^x &= 58 \\ \log_{10}(9^x) &= \log_{10}(58) \\ x \times \log_{10}(9) &= \log_{10}(58) \\ x &= \frac{\log_{10}(58)}{\log_{10}(9)} \\ x &= 1.85\end{aligned}$$

(d)

$$\begin{aligned}e^x &= 4 \\ \ln(e^x) &= \ln(4) \\ x \times \ln(e) &= \ln(4) \\ x &= 1.39\end{aligned}$$

(e)

$$\begin{aligned}2e^x &= 12 \\ e^x &= 6 \\ \ln(e^x) &= \ln(6) \\ x \times \ln(e) &= \ln(6) \\ x &= 1.79\end{aligned}$$

(f)

$$\begin{aligned}7e^x &= 21 \\ e^x &= 3 \\ \ln(e^x) &= \ln(3) \\ x \times \ln(e) &= \ln(3) \\ x &= 1.10\end{aligned}$$

(g)

$$\begin{aligned}5e^x + 4 &= 14 \\ 5e^x &= 10 \\ e^x &= 2 \\ \ln(e^x) &= \ln(2) \\ x \times \ln(e) &= \ln(2) \\ x &= 0.69\end{aligned}$$

(h)

$$\begin{aligned}3e^{2x+1} &= 15 \\e^{2x+1} &= 5 \\\ln(e^{2x+1}) &= \ln(5) \\(2x+1) \times \ln(e) &= \ln(5) \\2x+1 &= \ln(5) \\2x &= \ln(5) - 1 \\x &= \frac{\ln(5) - 1}{2} \\x &= 0.30\end{aligned}$$

(i)

$$\begin{aligned}8e^{5x-4} &= 32 \\e^{5x-4} &= 4 \\\ln(e^{5x-4}) &= \ln(4) \\(5x-4) \times \ln(e) &= \ln(4) \\5x-4 &= \ln(4) \\5x &= \ln(4) + 4 \\x &= \frac{\ln(4) + 4}{5} \\x &= 1.08\end{aligned}$$

(j)

$$\begin{aligned}6e^{9x+7} + 2 &= 20 \\6e^{9x+7} &= 18 \\e^{9x+7} &= 3 \\\ln(e^{9x+7}) &= \ln(3) \\(9x+7) \times \ln(e) &= \ln(3) \\9x+7 &= \ln(3) \\9x &= \ln(3) - 7 \\x &= \frac{\ln(3) - 7}{9} \\x &= -0.66\end{aligned}$$

5. (a)

$$\begin{aligned}\frac{x}{3} + \frac{4x}{5} &= \frac{1}{3} \\\frac{5x}{15} + \frac{12x}{15} &= \frac{1}{3} \\\frac{17x}{15} &= \frac{1}{3} \\\frac{17x}{15} \times 15 &= \frac{1}{3} \times 15 \\17x &= 5 \\x &= \frac{5}{17}\end{aligned}$$

(b)

$$\begin{aligned}\frac{15y}{4} - 5 &= \frac{y}{2} \\ \frac{15y}{4} - \frac{20}{4} &= \frac{2y}{4} \\ \frac{15y - 20}{4} &= \frac{2y}{4} \\ \frac{15y - 20}{4} \times 4 &= \frac{2y}{4} \times 4 \\ 15y - 20 &= 2y \\ 15y - 2y - 20 &= 2y - 2y \\ 13y - 20 &= 0 \\ 13y - 20 + 20 &= 0 + 20 \\ 13y &= 20 \\ \frac{13y}{13} &= \frac{20}{13} \\ y &= \frac{20}{13}\end{aligned}$$

(c)

$$\begin{aligned}\frac{2x - 4}{5} + 6 &= \frac{x}{2} \\ \frac{2(2x - 4)}{10} + \frac{60}{10} &= \frac{5x}{10} \\ \frac{4x - 8}{10} + \frac{60}{10} &= \frac{5x}{10} \\ \frac{4x + 52}{10} &= \frac{5x}{10} \\ \frac{4x + 52}{10} \times 10 &= \frac{5x}{10} \times 10 \\ 4x + 52 &= 5x \\ 4x - 5x + 52 &= 5x - 5x \\ -x + 52 &= 0 \\ -x + 52 - 52 &= 0 - 52 \\ -x &= -52 \\ x &= 52\end{aligned}$$

(d)

$$\begin{aligned}\frac{4h-1}{2} - \frac{2h+5}{3} &= 0 \\ \frac{3(4h-1)}{6} - \frac{2(2h+5)}{6} &= 0 \\ \frac{12h-3}{6} - \frac{4h+10}{6} &= 0 \\ \frac{8h-13}{6} &= 0 \\ \frac{8h-13}{6} \times 6 &= 0 \times 6 \\ 8h-13 &= 0 \\ 8h-13+13 &= 0+13 \\ 8h &= 13 \\ \frac{8h}{8} &= \frac{13}{8} \\ h &= \frac{13}{8}\end{aligned}$$